Technical report — Sovereign Systematic Review & Meta-Analysis Engine

A deeper write-up than the README: what you built, how it works, and what the numbers say.

- **Track:** `Track 2B — Sovereign Systematic Review & Meta-Analysis Engine`
- **Event:** Online
- **Team:** Hack Apertus participant team (identified in the submission portal)
- **Demo:** Submitted separately through the Hack Apertus submission portal

1. Summary

Clinical systematic reviews are bottlenecked by slow, error-prone extraction from trial PDFs. This prototype uses Apertus to propose PICO and risk-of-bias records, then independently verifies every evidence quote with PyMuPDF before exporting an auditable CSV and highlighted PDFs. In deterministic demo mode, all supplied sample claims are located and coordinate-locked without network access.

2. Architecture

Components and data flow are documented in [`docs/architecture.md`](docs/architecture.md).

Target architecture (mandatory)

State which of the three architectures your project is deployable in, and how it meets that constraint:

- **a) On-premise** — on the organisation's own infrastructure, under its own administration.
- **b) Air-gapped** — with no external network connection at runtime.
- **c) Sovereign Swiss cloud** — on a cloud platform operated in Switzerland, under Swiss jurisdiction, with Swiss data residency.

List any external dependencies, and separate build time from runtime.

3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-8B` through an OpenAI-compatible endpoint
- **How it is used:** inference | fine-tuning | evaluation | red-teaming | agents / tool use
- **Where it runs:** `local weights, hosted endpoint, ...`

The production adapter targets an OpenAI-compatible Apertus endpoint. Demo mode is deterministic and does not pretend to perform inference. Optional local weights are isolated in `requirements-local.txt` because model weights are too large and deployment-specific to commit.

4. Data

The checked-in sample trials are synthetic and generated at runtime. Real clinical PDFs must be mounted locally and are never uploaded by the application. No patient data or non-redistributable corpus is included. If data comes from human subjects or contains personal information, describe how consent was obtained.

5. Evaluation

How you measured success: task, metric, baseline.

Setup	Metric	Result
Setup	Metric	Result
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Demo pipeline	Quote verification	6/6 sample evidence anchors verified
Demo pipeline	Network dependency	None at runtime

6. Limitations

Where it breaks, what you did not test, and known failure modes.

7. Reproducibility

Run `make test` and `make demo` from `track_2b/`, or `make run` to start the Dockerized dashboard. API mode requires `APERTUS_API_KEY`, while demo mode requires no key or outbound network. Record the submission commit and the container image digest for formal evaluation.

8. Next steps

What you would build with another month.

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References